

- *4. Two different products, P1 and P2, can be manufactured by one or both of two different machines, M1 and M2. The unit processing time of either product on either machine is the same. The daily capacity of machine M1 is 200 units (of either P1 or P2, or a mixture of both) and the daily capacity of machine M2 is 250 units. The shop supervisor wants to balance the production schedule of the two machines such that the total number of units produced on one machine is within 5 units of the number produced on the other. The profit per unit of P1 is \$10 and that of P2 is \$15. Set up the problem as an LP in equation form.

x is within 5 units of y
 $|x - y| \leq 5$ or $|y - x| \leq 5$
 ~~x_1 : P1~~
 ~~x_2 : P2~~

x_{ij} = # of units product i produced in machine j .

x_{11} = # of units P1 produced in machine 1.
 x_{12} = " " " " " " 2.
 x_{21} = " " " P2 " " 1
 x_{22} = " " " " " " 2

$$\max z = 10(x_{11} + x_{12}) + 15(x_{21} + x_{22})$$

$$x_{11} + x_{21} \leq 200$$

$$x_{12} + x_{22} \leq 250$$

$$|x_{11} + x_{21} - x_{12} - x_{22}| \leq 5$$

$$x_{11}, x_{21}, x_{12}, x_{22} \geq 0$$

$$x_{11} + x_{21} - x_{12} - x_{22} \leq 5$$

$$x_{11} + x_{21} - x_{12} - x_{22} \geq -5$$

$$-x_{11} - x_{21} + x_{12} + x_{22} \leq 5$$

2. Two products are manufactured in a machining center. The production times per unit of products 1 and 2 are 10 and 12 minutes, respectively. The total regular machine time is 2500 minutes per day. In any one day, the manufacturer can produce between 150 and 200 units of product 1, but no more than 45 units of product 2. Overtime may be used to meet the demand at an additional cost of \$.50 per minute. Assuming that the unit profits for products 1 and 2 are \$6.00 and \$7.50, respectively, formulate the problem as an LP model, then solve with TORA, Solver, or AMPL to determine the optimum production level for each product as well as any overtime needed in the center.

10 x_1 = # of units for product 1. (per day)
 12 x_2 = " " " " " 2. (")

2500 = total machine time.

$$10x_1 + 12x_2 \leq 2500$$

$$150 \leq x_1 \leq 200$$

$$x_2 \leq 45$$

- (1) overtime occurs $2500 \uparrow$ $2600?$ ~~$\times$~~
 (2) overtime does not occur $2500 \downarrow$ $2000.$ $\times$.
 additional cost $\leftarrow$

$$10x_1 + 12x_2 + x_3 = 2500,$$

x_3 : free.

$x_3 = 500 > 0$, $10x_1 + 12x_2 = 2000$ (case 2)
 $x_3^- = \text{unused minutes}$ no overtime needed.

$x_3 = -100 < 0$, $10x_1 + 12x_2 = 2600$ (case 1)
 $x_3^+ = \text{overtime minutes} = 100$ overtime needed.

$$x_3 = x_3^- - x_3^+; \quad x_3^-, x_3^+ \geq 0$$

$$x_3 = 500 - 0; \quad x_3^- = 500, x_3^+ = 0$$

$$x_3 = 0 - 100; \quad x_3^- = 0, x_3^+ = 100$$

$$10x_1 + 12x_2 + x_3^- - x_3^+ = 2500.$$

$$\max Z = \underbrace{6x_1 + 7.5x_2}_{\text{min. Profit.}} + \underbrace{0.5x_3^+}_{\text{Cost.}}$$

$\max(\text{cost}) \Rightarrow$ increase the cost.

$\max(-\text{cost}) \Rightarrow$ decrease the cost.

$$\max f = \min(-f)$$

$$\min f = \max(-f)$$

$$\max f(x) = x^2$$

$$\min -f(x) = -x^2$$

$$f(x) = x^2$$

$$f(x) = -x^2$$

$$\begin{aligned} \max z &= 6x_1 + 7.5x_2 - 0.5x_3^+ \\ \text{s.t.} \quad 10x_1 + 12x_2 + x_3^- - x_3^+ &= 2500 \\ x_1 &\leq 200 \\ x_1 &\geq 150 \\ x_2 &\leq 45 \\ x_1, x_2, x_3^-, x_3^+ &\geq 0. \end{aligned}$$

$$\begin{aligned} x_1 &= 200 \\ x_2 &= 45 \\ x_3^+ &= 40 \\ z &= \$1517.50 \end{aligned}$$

$$\begin{aligned} \text{Maximize } z &= 2x_1 + 3x_2 \\ \text{subject to } 2x_1 + x_2 &\leq 4 \\ x_1 + 2x_2 &\leq 5 \\ x_1, x_2 &\geq 0 \end{aligned}$$

$\Rightarrow$

$$\begin{aligned} \max z &= 2x_1 + 3x_2 \\ 2x_1 + x_2 + s_1 &= 4 \\ x_1 + 2x_2 + s_2 &= 5 \\ x_1, x_2, s_1, s_2 &\geq 0. \end{aligned}$$

Iteration 1

Basic $\Rightarrow (s_1, s_2)$
Nonbasic $\Rightarrow x_1, x_2$

$$s_1 = 4 - 2x_1 - x_2$$

$$s_2 = 5 - x_1 - 2x_2$$

$$z = 2x_1 + 3x_2$$

$$z = 2x_1 + 3x_2 = 0$$

$$x_1 = x_2 = 0 \Rightarrow s_1 = 4 \\ s_2 = 5$$

Limiting value
4

$$5/2$$

$$\min(4, 5/2) = \frac{5}{2}$$

$$s_2 = 5 - 0 - 2 \cdot 4 = -3 \quad \times$$

x_2 : Entering variable.

$$\cancel{x_2 = 10}, x_1 = 0.$$

$$s_1 = 4 - 2 \cdot 0 - 10 = -6. \text{ (infeasible)}$$

$$\cancel{x_2 = 8}, x_1 = 0, s_1 = 4 - 2 \cdot 0 - 8 = -4.$$

$$s_1 = 4 - x_2 = 0.$$

$$x_2 = \frac{5}{2} \Rightarrow s_1 = 4 - \frac{5}{2} = \frac{3}{2}$$

$$S_1 = 4 - x_2 = 0.$$

$$x_2 = 4.$$

$$x_2 = \frac{3}{2} \Rightarrow S_1 = 4 - \frac{3}{2} = \frac{5}{2}$$

$$\leftarrow S_2 = 5 - 2 \cdot \frac{5}{2} = 0.$$

it will become nonbasic.

Iteration 2

$S_1, \cancel{S_2}, x_2$

Basic : S_1, x_2

Nonbasic = x_1, S_2 .

$$x_2 = \frac{5}{2}, x_1 = 0, S_2 = 0,$$

$$S_1 = \frac{3}{2},$$

$$z = 3 \cdot \frac{5}{2} = \frac{15}{2}$$

$$S_1 = 4 - 2x_1 - x_2$$

$$S_2 = 5 - x_1 - 2x_2$$

$$z = 2x_1 + 3x_2$$

$$\rightarrow 2x_2 = 5 - x_1 - S_2$$

$$x_2 = \frac{5}{2} - \frac{1}{2}x_1 - \frac{1}{2}S_2$$

$$S_1 = 4 - 2x_1 - \left(\frac{5}{2} - \frac{1}{2}x_1 - \frac{1}{2}S_2 \right)$$

$$\frac{5}{2} - \frac{1}{2}x_1 = 0$$

$x_1 = 5$

$$= 4 - 2x_1 - \frac{5}{2} + \frac{1}{2}x_1 + \frac{1}{2}S_2$$

$$S_1 = \frac{3}{2} - \frac{3}{2}x_1 + \frac{1}{2}S_2$$

$$x_1 = 1$$

$$z = 2x_1 + 3 \left(\frac{5}{2} - \frac{1}{2}x_1 - \frac{1}{2}S_2 \right)$$

$$z = \frac{15}{2} + \frac{1}{2}x_1 - \frac{3}{2}S_2$$

$$x_1 \uparrow ?$$

$$\text{or } S_2 \uparrow ?$$

to increase z - ?

x_1 : Entering variable.

x_1 : Entering variable.

Limiting values from x_2 equation = 5
" " " S_1 " = 1

$$\min(1, 5) = 1.$$

entering -
 $x_1 = 1$

Leaving variable = S_1 .

$$x_2 = 2$$

$S_1 = 0 \Rightarrow$ becomes nonbasic.

Iteration 3

Basic = x_1, x_2 .

Nonbasic = S_1, S_2 .

$$\left. \begin{array}{l} x_1 = 1 \dots \\ x_2 = 2 \dots \end{array} \right\} \text{in terms of } S_1, S_2.$$

$$S_1 = \frac{3}{2} - \frac{3}{2}x_1 + \frac{1}{2}S_2$$

$$2 = 2 \cdot 1 + 3 \cdot 2 = 8.$$

$$\frac{3}{2}x_1 = \frac{3}{2} - S_1 + \frac{1}{2}S_2$$

$$x_1 = 1 - \frac{2}{3}S_1 + \frac{1}{3}S_2$$

$$x_2 = \frac{5}{2} - \frac{1}{2}\left(1 - \frac{2}{3}S_1 + \frac{1}{3}S_2\right) - \frac{1}{2}S_2$$

$$= \frac{5}{2} - \frac{1}{2} + \frac{2}{3}S_1 - \frac{1}{6}S_2 - \frac{1}{2}S_2$$

$$x_2 = 2 + \frac{2}{3}S_1 - \frac{2}{3}S_2$$

$$z = \frac{15}{2} + \frac{1}{2} \left(1 - \frac{2}{3}S_1 + \frac{1}{3}S_2 \right) - \frac{3}{2}S_2$$

$$z = 8 - \frac{1}{3}S_1 - \frac{4}{3}S_2$$

$$-\frac{1}{3} \quad +\frac{1}{6} - \frac{8}{6}$$

$$-\frac{8}{6}$$

Entering variable = ?

$S_1 \uparrow \quad z \downarrow \quad S_2 \uparrow \quad z \downarrow \quad \max z.$

$$z = 0 \implies z = \frac{15}{2} \implies z^* = 8$$

No entering variable
STOP.

(A)

(B)

(C)

$z - 2x_1 - 3x_2 = 0$ the most negative

RHS
coefficient
in
the E.V. column.

Basic	x_1	x_2 E.V.	S_1	S_2	Solution RHS.
z	-2	-3	0	0	0
S_1	2	1	1	0	4
S_2 L.V.	1	2	0	1	5
z	$-\frac{1}{2}$	0	0	$\frac{3}{2}$	$\frac{15}{2}$
S_1 L.V.	$\frac{3}{2}$	0	1	$-\frac{1}{2}$	$\frac{3}{2}$
x_2	$\frac{1}{2}$	1	0	$\frac{1}{2}$	$\frac{5}{2}$

z -row. (-)
min Ratio

$$4/1 = 4$$

$$5/2 = \frac{5}{2}$$

min ratio

$$\frac{3/2}{3/2} = 1 \implies$$

$$\frac{5/2}{1/2} = 5$$

$$\min(1, 5) = 1$$

Multiply by (-) to zero.

multiply by (-1) to zero.

$$\min(1, 5) = 1 //$$

$$\begin{array}{rcccccc} & 2 & 1 & 1 & 0 & 4 \\ + & -\frac{1}{2} & -1 & 0 & -\frac{1}{2} & \frac{5}{2} \\ \hline & \frac{3}{2} & 0 & 1 & -\frac{1}{2} & \frac{3}{2} \end{array}$$

$$\text{Old } z\text{-row} \quad (-3) + (\text{New } x_2\text{-row}) \times (3) = (\text{New } z\text{-row})$$

$$\begin{array}{rcccccc} \text{Old } z\text{-row} & -2 & -3 & 0 & 0 & 0 \\ 3 \times \text{New } x_2\text{-row} & \frac{3}{2} & 3 & 0 & \frac{3}{2} & \frac{15}{2} \\ + & & & & & \\ \hline & -\frac{1}{2} & 0 & 0 & \frac{3}{2} & \frac{15}{2} \end{array}$$

z	$-\frac{1}{2}$	0	0	$\frac{3}{2}$	$\frac{15}{2}$	
s_1	$\frac{3}{2}$	0	1	$-\frac{1}{2}$	$\frac{3}{2}$	
x_2	$\frac{1}{2}$	1	0	$\frac{1}{2}$	$\frac{5}{2}$	$x_1 \longleftrightarrow s_1$
z	0	0	$\frac{1}{3}$	$\frac{4}{3}$	8	
x_1	1	0	$\frac{2}{3}$	$-\frac{1}{3}$	1	$\frac{2}{3}$
x_2	0	1	$-\frac{1}{3}$	$\frac{2}{3}$	2	$\frac{9}{6} - \frac{1}{6}$

$E.V = ?$
 $z + 0x_1 + 0x_2 + \frac{1}{3}s_1 + \frac{4}{3}s_2 = 8$

$$E.V = ?$$

$$z + 0x_1 + 0x_2 + \frac{1}{3}s_1 + \frac{4}{3}s_2 = 8.$$

$$\frac{1}{2}(\text{New } x_1\text{-row}) = \begin{array}{ccccc} -\frac{1}{2} & 0 & -\frac{1}{3} & \frac{1}{6} & -\frac{1}{2} \end{array}$$

$$O.L. \ x_2\text{-row} = \begin{array}{ccccc} \frac{1}{2} & 1 & 0 & \frac{1}{2} & \frac{5}{2} \end{array}$$

$$\begin{array}{ccccc} 0 & 1 & -\frac{1}{3} & \frac{2}{3} & 2. \end{array}$$

$$\frac{1}{2}(\text{new } x_1\text{row}) + O.L. \ z\text{-row} = \text{New } z\text{-row}.$$

Since there exists no negative values in $z\text{-row}$, we reached the optimum.

$$x_1^* = 1, \quad x_2^* = 2,$$

$$z^* = 8$$

$$s_1^* = 0, \quad s_2^* = 0.$$